

LIE ALGEBRAS

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1. BASIC DEFINITIONS AND EXAMPLES

Definition 1.1. An *algebra* is a vector space over a field $\mathbb{F}$ endowed with a binary operation that is bilinear

$$\begin{aligned}a(\lambda b + \mu c) &= \lambda ab + \mu ac \\(\lambda b + \mu c)a &= \lambda ba + \mu ca\end{aligned}$$

Definition 1.2. A *Lie algebra* is an algebra $\mathfrak{g}$ with a product $[\cdot, \cdot]$ we call *bracket* that satisfies

- (1) $[x, x] = 0 \quad \forall x \in \mathfrak{g},$
- (2) (Jacobi identity.)

Definition 1.3. A *simple Lie algebra* is a Lie algebra that has no nontrivial proper ideals.

As a rule of thumb I keep in mind the following silly computation:

$$\log \det A = \log \prod_i \lambda_i = \sum \log \lambda_i = \text{“trlog } A\text{”}$$

And recall that exponent map goes from $T_e G = \mathfrak{g} \rightarrow G$, so that logarithm would go from $G \rightarrow \mathfrak{g}$. This is why I remember that the condition on a classical Lie group of *having determinant 1* goes to *having vanishing trace* in the Lie algebra. (Because $\log 1 = 0$.)

Example 1.4. (1) The *special linear Lie algebra*

$$\mathfrak{sl}_n = \{\text{Mat}_n | \text{Tr}(A) = 0\}$$

which is just obvious from the slogan above.

(2) The *special orthogonal Lie algebra*

$$\mathfrak{so}_n = \{A \in \text{Mat}_n \mid A + A^T = 0\}$$

which is obvious from: $\text{SO}(n)$ = isometries, so $\langle v, v \rangle = \langle Av, Av \rangle = \langle v, A^T Av \rangle$, so $\text{SO}(n) = \{A \in \text{Mat}_n : A^{-1} = A^T\}$, and then $0 = \log 1 = \log(AA^T) = \log A + \log A^T$.

(3) The *symplectic Lie algebra*

$$\mathfrak{sp}_{2n} = \{A \in \text{Mat}_{2n} : \Omega A + A^T \Omega = 0\}$$

$$\text{where } \Omega = \begin{pmatrix} 0 & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix}.$$

Pending. Why this makes sense?

Example 1.5. If A is an associative Lie algebra, then A with the bracket $[a, b] = ab - ba$ is a Lie algebra, denoted by A_- . (It is an exercise to verify Jacobi's identity.) This gives $\mathfrak{gl}_V := \text{End}(V)_-$.

Definition 1.6. A *Lie subalgebra* is a vector subspace $\mathfrak{h} \subset \mathfrak{g}$ such that $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$.

Definition 1.7. A subspace $\mathfrak{h} \subset \mathfrak{g}$ is called an *ideal* if $[\mathfrak{h}, \mathfrak{g}] \subset \mathfrak{h}$.

Recall that a bilinear form $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ is *invariant* if $([a, b], c) = (a, [b, c])$ for all $a, b, c \in \mathfrak{g}$.

Definition 1.8. For any finite dimensional Lie algebra $\mathfrak{g}$ and $x \in \mathfrak{g}$ we write

$$\text{ad}_x = [x, -]$$

Exercise 1.9. Show that the map $\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$, $x \mapsto \text{ad}_x$ is a representation, i.e. a morphism of Lie algebras.

Definition 1.10. The *Killing form* is

$$\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}_x \text{ad}_y$$

Exercise 1.11. Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.

Proof. (David.) It's enough to show that $\mathfrak{g}$ is *perfect*, i.e. that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. In this case, let $a, b \in \mathfrak{g}$ and suppose that $b = [x, y]$. Then

$$\begin{aligned} (a, b) &= (a, [x, y]) = (a, -[y, x]) = (-[a, y], x) = ([y, a], x) \\ &= (y, [a, x]) = (y, -[x, a]) = (-[y, x], a) = ([x, y], a) = (b, a) \end{aligned}$$

To confirm that $\mathfrak{g}$ is perfect just observe that $[\mathfrak{g}, \mathfrak{g}]$ is a nontrivial ideal of $\mathfrak{g}$. □

Definition 1.12. A *semisimple Lie algebra* is a direct sum of simple ones

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \dots \oplus \mathfrak{g}_s$$

Theorem 1.13 (Cartan). *The finite dimensional Lie algebra $\mathfrak{g}$ is semisimple if and only if its Killing form is nondegenerate.*

2. NILPOTENT AND SOLVABLE LIE ALGEBRAS

Let $\mathfrak{g}$ be a Lie algebra. Define

$$\mathfrak{g}^1 := [\mathfrak{g}, \mathfrak{g}], \quad \mathfrak{g}^2 := [\mathfrak{g}, \mathfrak{g}^1], \quad \dots, \quad \mathfrak{g}^n := [\mathfrak{g}, \mathfrak{g}^{n-1}]$$

and

$$\mathfrak{g}^{(1)} := [\mathfrak{g}, \mathfrak{g}], \quad \mathfrak{g}^{(2)} := [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}], \dots, \mathfrak{g}^{(n)} := [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}].$$

Definition 2.1. A Lie algebra $\mathfrak{g}$ is

- *nilpotent* if there exists n such that $\mathfrak{g}^n = 0$,
- *solvable* if there exists n such that $\mathfrak{g}^{(n)} = 0$.

3. WEIGHT SPACES AND LIE'S THEOREM

Lie's theorem says that solvable Lie algebras over closed fields of characteristic not zero have weights.

Definition 3.1. Let $\mathfrak{h}$ be a Lie algebra, $\pi : \mathfrak{h} \rightarrow \mathfrak{gl}_V$ a representation of $\mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$. The *weight space* of $\mathfrak{h}$ attached to λ is

$$V_\lambda^\mathfrak{h} := \{v \in V \mid \pi(h)v = \lambda(h)v, \forall h \in \mathfrak{h}\}.$$

If $V_\lambda^\mathfrak{h} \neq 0$, we say that λ is a *weight* for π .

Theorem 3.2 (Lie's theorem). *Let $\mathfrak{g}$ be a solvable Lie algebra and π a representation of $\mathfrak{g}$ on a finite dimensional vector space $V \neq 0$, over an algebraically closed field $\mathbb{F}$ of characteristic 0. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for π (that is, $V_\lambda^\mathfrak{g} \neq \{0\}$).*

Exercise 3.3. Show the following two corollaries of Lie's theorem:

- for all representations π of a solvable Lie algebra $\mathfrak{g}$ on a finite dimensional vector space V over an algebraically closed field $\mathbb{F}$, $\text{char} \mathbb{F} = 0$, there exists a basis for V for which the matrices of $\pi(\mathfrak{g})$ are upper triangular;
- a solvable subalgebra $\mathfrak{g} \subset \mathfrak{gl}_V$, (where V is finite-dimensional over an algebraically closed field $\mathbb{F}$, $\text{char} \mathbb{F} = 0$), is contained in the subalgebra of upper triangular matrices over $\mathbb{F}$ for some basis of V .

4. GENERALIZED EIGENSPACES AND GENERALIZED WEIGHT SPACES

Definition 4.1. A *generalized eigenspace* of $A \in \text{End}(V)$ with eigenvalue λ is

$$V_\lambda = \{v \in V : (A - \lambda \text{Id})^N = 0 \text{ for some positive integer } N\}.$$

It turns out that any linear operator on an algebraically closed field gives a decomposition into generalized eigenspaces via Jordan canonical form:

Proposition 4.2. *Let A be a linear operator on a finite-dimensional vector space V over an algebraically closed field $\mathbb{F}$, and let $\lambda_1, \dots, \lambda_s$ be all eigenvalues of A , and $n_1, \dots, n_s$ their multiplicities. Then one has the generalized eigenspace decomposition:*

$$V = \bigoplus_{i=1}^s V_{\lambda_i}, \quad \dim V_{\lambda_i} = n_i.$$

In particular, for a Lie algebra $\mathfrak{g}$ with a representation (i.e. Lie algebra morphism) $\pi : \mathfrak{g} \rightarrow \text{End}(V)$ we have

$$V = \bigoplus_{\lambda \in \mathbb{F}} V_{\lambda}^a, \quad V_{\lambda}^a = \{v \in V : (\pi(a)v - \lambda \text{Id})^N v = 0 \text{ for some } N \in \mathbb{N}\}$$

And even more particularly, for the adjoint representation we have

$$V = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_{\lambda}^a, \quad \mathfrak{g}_{\lambda}^a = \{b \in \mathfrak{g} : ([a, \cdot] - \alpha \text{Id})^N b = 0, \text{ for some } N \in \mathbb{N}\}$$

And a little more generally we have

Definition 4.3. Let $\mathfrak{h}$ be a Lie algebra with a representation π on a vector space V , and $\lambda \in \mathfrak{h}^*$. A *generalized weight space of $\mathfrak{h}$ in V attached to λ* is

$$V_{\lambda}^{\mathfrak{h}} = \left\{ v \in V : (\pi(a) - \lambda(a)\text{Id})^N v = 0, \begin{array}{l} \text{for some } N \in \mathbb{N} \\ \text{depending on } a \in \mathfrak{h} \\ \text{for all } a \in \mathfrak{h} \end{array} \right\}$$

Under the right conditions, a nilpotent subalgebra $\mathfrak{h} \subset \mathfrak{g}$ permits decomposing V as a direct sum of generalized weight spaces of $\mathfrak{h}$. Namely,

Theorem 4.4. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and π its representation on a finite-dimensional vector space V , over an algebraically closed field $\mathbb{F}$ of characteristic 0. Let $\mathfrak{h}$ be a nilpotent subalgebra of $\mathfrak{g}$. Then the following equalities hold:*

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_{\lambda}^{\mathfrak{h}}$$

$$(4.4.1) \quad \pi(\mathfrak{g}_{\alpha}^{\mathfrak{h}}) V_{\lambda}^{\mathfrak{h}} \subseteq V_{\lambda+\alpha}^{\mathfrak{h}}$$

Which in the case of the adjoint representation, looks like:

Definition 4.5. The *generalized root space decomposition* of a Lie algebra $\mathfrak{g}$ over an algebraically closed field $\mathbb{F}$ of characteristic zero with respect to a nilpotent subalgebra $\mathfrak{h}$ is the generalized weight space decomposition with respect to the adjoint representation. That is,

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha}^{\mathfrak{h}}$$

and it has the property that

$$(4.5.1) \quad [\mathfrak{g}_{\alpha}^{\mathfrak{h}}, \mathfrak{g}_{\beta}^{\mathfrak{h}}] \subseteq \mathfrak{g}_{\alpha+\beta}^{\mathfrak{h}}$$

It is important to make the distinction between the generalized weight space decomposition and the generalized root space decomposition; “we will see its convenience in later lectures, as we try to better understand the functions α appearing in the decomposition”.

Exercise 4.6. Take $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F})$ and $\mathfrak{h} = \{\text{diagonal matrices}\}$. Find the generalized weight space decomposition in both the tautological and the adjoint representations, and check the inclusions 4.4.1 and 4.5.1.

5. ZARISKY TOPOLOGY AND REGULAR ELEMENTS

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra of dimension d . For every element $a \in \mathfrak{g}$ the characteristic polynomial of ad_a must be of the form

$$\det(\text{ad}_a - \lambda \text{Id}) = (-\lambda)^d + c_{d-1}(-\lambda)^{d-1} + \dots + \det(\text{ad}_a).$$

[to be honest I don't see why the constant term must be the determinant of ad_a , but OK...] since $[a, a] = 0$, $\det(\text{ad}_a) = 0$, i.e., the constant term vanishes.

Exercise 5.1. Show that c_j is a homogeneous polynomial on $\mathfrak{g}$ of degree $d - j$.

Proof. The term “polynomial on $\mathfrak{g}$ ” is not clear. But we follow [?] to interpret this is only as being a polynomial on the a^i where $a = a^i X_i$ for some basis X_i of $\mathfrak{g}$.

Then we just notice that denoting $[X_i, X_j] := L_{ij}$ we have

$$[a, X_j] = [a^i X_i, X_j] = a^i [X_i, X_j] = a^i L_{ij}$$

which is a linear combination of the a^i . Then the matrix representation of ad_a in terms of the basis X_i is

$$[\text{ad}_a] = \begin{pmatrix} a^i L_{i1}^1 & \dots & a^i L_{in}^1 \\ \vdots & & \vdots \\ a^i L_{i1}^n & \dots & a^i L_{in}^n \end{pmatrix}$$

subtracting λId and taking determinant we obtain that a term with λ^k would have for coefficient a product of k of the linear combinations $a^i L_{i\ell}^j$ for varying j and ℓ . Such a product is understood to be a homogeneous polynomial of degree k in $\mathfrak{g}$. $\square$

Definition 5.2. I think this goes for any Lie algebra $\mathfrak{g}$:

- The *rank* of $\mathfrak{g}$ is the smallest integer r such that $c_r(a)$ is not the zero polynomial on $\mathfrak{g}$.
- An element $a \in \mathfrak{g}$ is called *regular* if $c_r(a) \neq 0$.
- The *discriminant* of $\mathfrak{g}$ is the nonzero polynomial $c_r(a)$ of degree $d - r$, what?

Explanation: we compute the polynomial characteristic w.r.t. ad_a for every $a \in \mathfrak{g}$. Express this polynomial as a polynomial with coefficients in $\mathfrak{g}[t]$, (here's the question: this polynomial seems to me to be a polynomial in $\mathbb{F}$.) This polynomial does not have a constant term. But what is the next smallest-degree monomial? 1? 2? That's the rank of the Lie algebra.

6. CARTAN SUBALGEBRA

Recall that $\mathfrak{h} \subset \mathfrak{g}$ is an ideal if $[a, \mathfrak{h}] \subset \mathfrak{h}$ for all $a \in \mathfrak{g}$. Maybe $\mathfrak{h}$ is not an ideal, but we can consider the largest subalgebra of $\mathfrak{g}$ where $\mathfrak{h}$ is an ideal. This is called the normalizer.

Definition 6.1. Let $\mathfrak{h}$ be a subalgebra of a Lie algebra $\mathfrak{g}$. The *normalizer* of $\mathfrak{h}$ is $N_{\mathfrak{g}}(\mathfrak{h}) := \{a \in \mathfrak{g} \mid [a, \mathfrak{h}] \subset \mathfrak{h}\}$

I think behind the Cartan subalgebra is that, roughly, taking product with anything not in $\mathfrak{h}$ leaves $\mathfrak{h}$.

Definition 6.2. A *Cartan subalgebra* of a Lie algebra $\mathfrak{g}$ is a subalgebra $\mathfrak{h}$, satisfying the following two conditions:

- (1) $\mathfrak{h}$ is a nilpotent Lie algebra,

$$(2) \ N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

Proposition 6.3. *Let $\mathfrak{g} \subset \mathfrak{gl}_n(\mathbb{F})$ be a subalgebra containing a diagonal matrix $a = \text{diag}(a_1, \dots, a_n)$ with distinct a_i , and let $\mathfrak{h}$ be the subspace of all diagonal matrices in $\mathfrak{g}$. Then $\mathfrak{h}$ is a Cartan subalgebra (of $\mathfrak{g}$ I guess).*

Idea of proof. Consider the following illustrative example:

$$\begin{aligned} & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} - \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} a\lambda_1 & b\lambda_2 \\ c\lambda_1 & d\lambda_2 \end{pmatrix} - \begin{pmatrix} \lambda_1 a & \lambda_1 b \\ \lambda_2 c & \lambda_2 d \end{pmatrix} \\ &= \begin{pmatrix} 0 & b\lambda_2 - \lambda_1 b \\ \lambda_1 c - \lambda_2 c & 0 \end{pmatrix} \end{aligned}$$

which says that any non-diagonal matrix would escape $\mathfrak{h}$. $\square$

Theorem 6.4 (Cartan). *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over an algebraically closed field $\mathbb{F}$. Let $a \in \mathfrak{g}$ be a regular element (which exists since $\mathbb{F}$ is infinite), and let $\mathfrak{g} = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_{\lambda}^a$ be the generalized eigenspace decomposition of $\mathfrak{g}$ with respect to ad_a . Then $\mathfrak{g}_0^a$ is a Cartan subalgebra.*

Idea of proof. First recall what is $\mathfrak{h} = \mathfrak{g}_0^a$, the set of elements $b \in \mathfrak{g}$ such that $\text{ad}_a^N(b) = 0$. $\square$

Remark 6.5. The rank of a Lie algebra is the dimension of $\mathfrak{g}_0^a = \mathfrak{h}$.

Proposition 6.6. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over an algebraically closed field $\mathbb{F}$ of characteristic zero and let $\mathfrak{h} \subset \mathfrak{g}$ be a Cartan subalgebra. Then $\mathfrak{g}_0 = \mathfrak{h}$ in the generalized weight space decomposition.*

Idea of proof. Engel's theorem. . . $\square$

7. SEMISIMPLE LIE ALGEBRAS

Definition 7.1. A *radical* $R(\mathfrak{g})$ of a finite-dimensional Lie algebra $\mathfrak{g}$ is a solvable ideal of $\mathfrak{g}$ of maximal possible dimension.

Proposition 7.2. *The radical ideal of $\mathfrak{g}$ contains any solvable ideal of $\mathfrak{g}$ and is unique.*

If $\mathfrak{g}$ is a finite dimensional solvable Lie algebra, then $R(\mathfrak{g}) = \mathfrak{g}$. The opposite case is when $R(\mathfrak{g}) = 0$.

Definition 7.3. A finite-dimensional Lie algebra $\mathfrak{g}$ is called *semisimple* if $R(\mathfrak{g}) = 0$.

In [?, Lecture 11] we have the tools needed for a characterization of semisimple Lie algebras in terms of the Killing form:

Theorem 7.4 (Cartan). *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field of characteristic 0. Then the Killing form on $\mathfrak{g}$ is non-degenerate if and only if $\mathfrak{g}$ is semisimple. Moreover, if $\mathfrak{g}$ is semisimple and $\mathfrak{a} \subset \mathfrak{g}$ is an ideal, then the restriction of the Killing form to $\mathfrak{a}$, $K|_{\mathfrak{a} \times \mathfrak{a}}$, is also nondegenerate and coincides with the Killing form of $\mathfrak{a}$.*

REFERENCES